Stability and Scalability in Mokken Scale Analysis: Challenges of Small Samples and Deviant Items

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Introduction

- Mokken Scale Analysis (MSA) is frequently used in social sciences; i.e. children with special needs
- Particularly, when parametric assumptions are not met (Sijtsma & Molenaar, 2002)
- Minimum sample size requirements seem high (Straat et al., 2014; Watson et al., 2018)

→ Performance of MSA remains underexplored

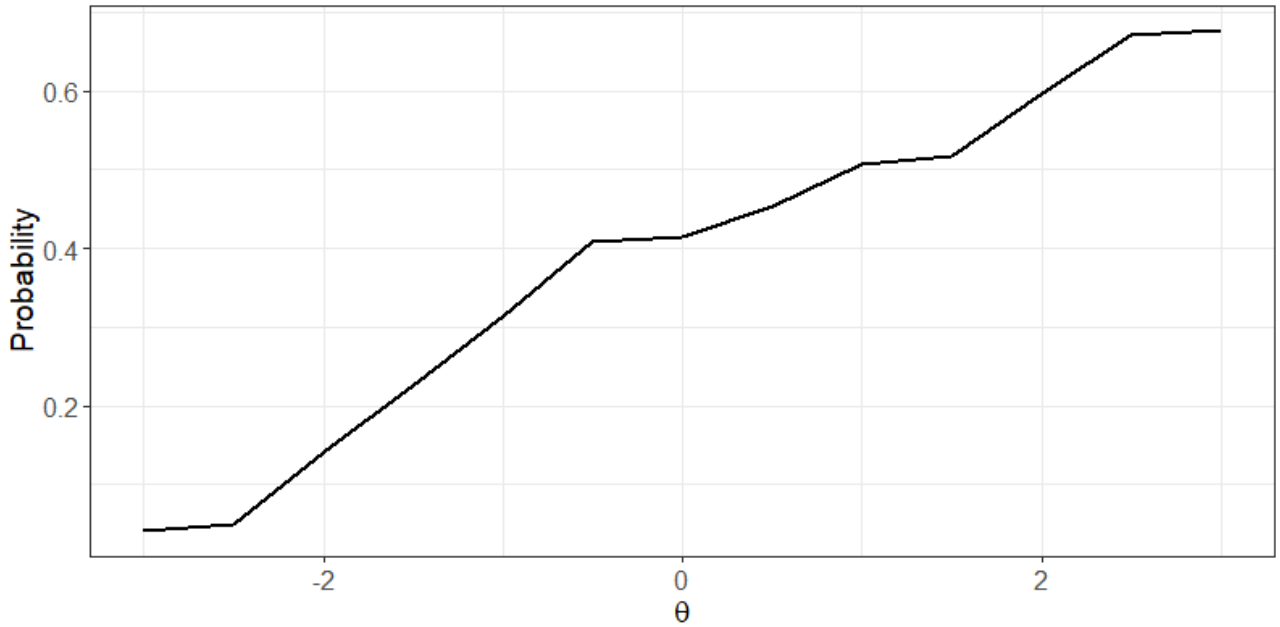
- in small samples
- with dichotomous data

Research Question

- *How does confirmatory and exploratory MSA perform with dichotomous data in relation to (small) sample size?*
  - *measured by H coefficients and the capability of the automated item search procedure to retrieve complete or unidimensional scales,*
  - *under varying conditions of item discrimination, item difficulty patterns, lower bound criteria, and the presence of deviant items?*

Monotone Homogeneity Model (*MHM*)

- *MHM* is the basis of *MSA* (Mokken, 1971)
- Non–parametric item response model (*IRT*) (Mokken, 1971)
- Sum score on items orders people on the latent trait (Sijtsma & Molenaar, 2002)



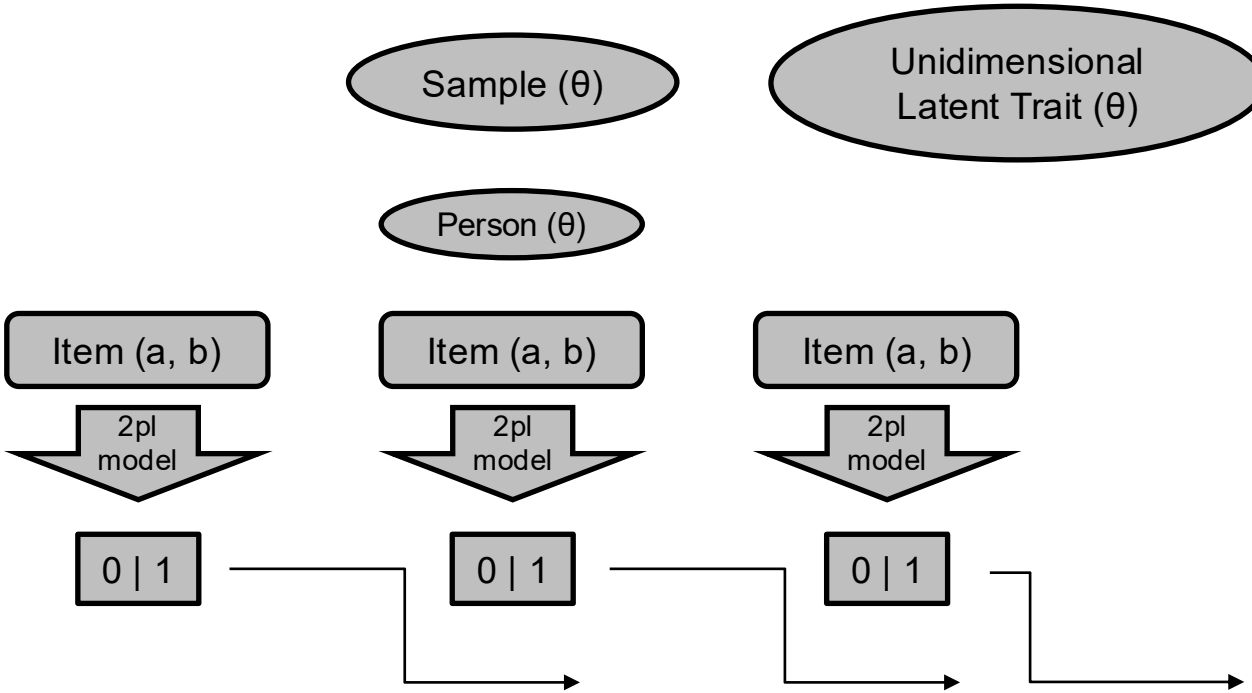
Methods: Mokken Scaling

- Scalability coefficients

$$H_{jk} = \frac{Cov(X_j, X_k)}{Cov(X_j, X_k)_{max}} \qquad H_j = \frac{Cov(X_j, R_{-j})}{Cov(X_j, R_{-j})_{max}} \qquad H = \frac{\sum_{j=1}^{J} Cov(X_j, R_{-j})}{\sum_{j=1}^{J} Cov(X_j, R_{-j})_{max}}$$

- Items form a Mokken scale if
  - all H_{jk} are > 0
  - all $H_j \geq$ lower bound $c \geq 0$ (Mokken, 1971)
- Automated Item Search Procedure (AISP)

Methods: Simulation Design

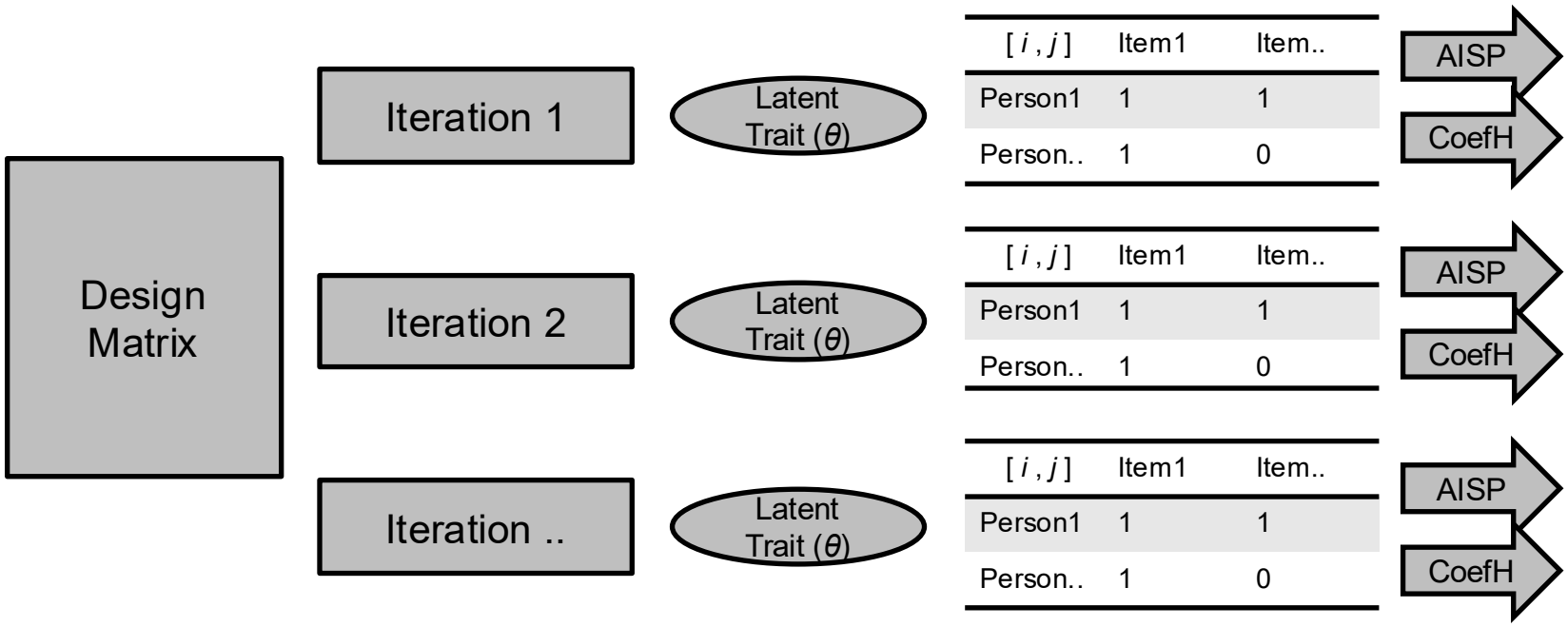


[i , j]	Item1	Item2	Item..
Person1	1	1	0
Person2	1	0	1
Person..	1	1	0

Methods: Simulation Design

Parameter	Primary Simulation
Latent Trait	$\theta \sim N(0,1)$
Test Length	$J = 10, 20$
Sample Size	$N = 50, 75, \ldots, 225, 250$
Item Discrimination	$a = 1.0, 1.3, 1.6, 4.0$
Lower Bound	$c = .2, .25, .3, .35, .4$
Item Difficulty Pattern	B = equal, normal, cluster
Iterations	*iter* = 500

Methods: Simulation Design



Methods: Outcome Measures

- Confirmatory MSA
  - *Distribution of H^* coefficient
- Exploratory MSA
  - AISP retrieval rate of *unidimensional* scale solution

Results

Figure 1. *H Coefficient versus Sample Size for Different Levels of Item Discrimination*

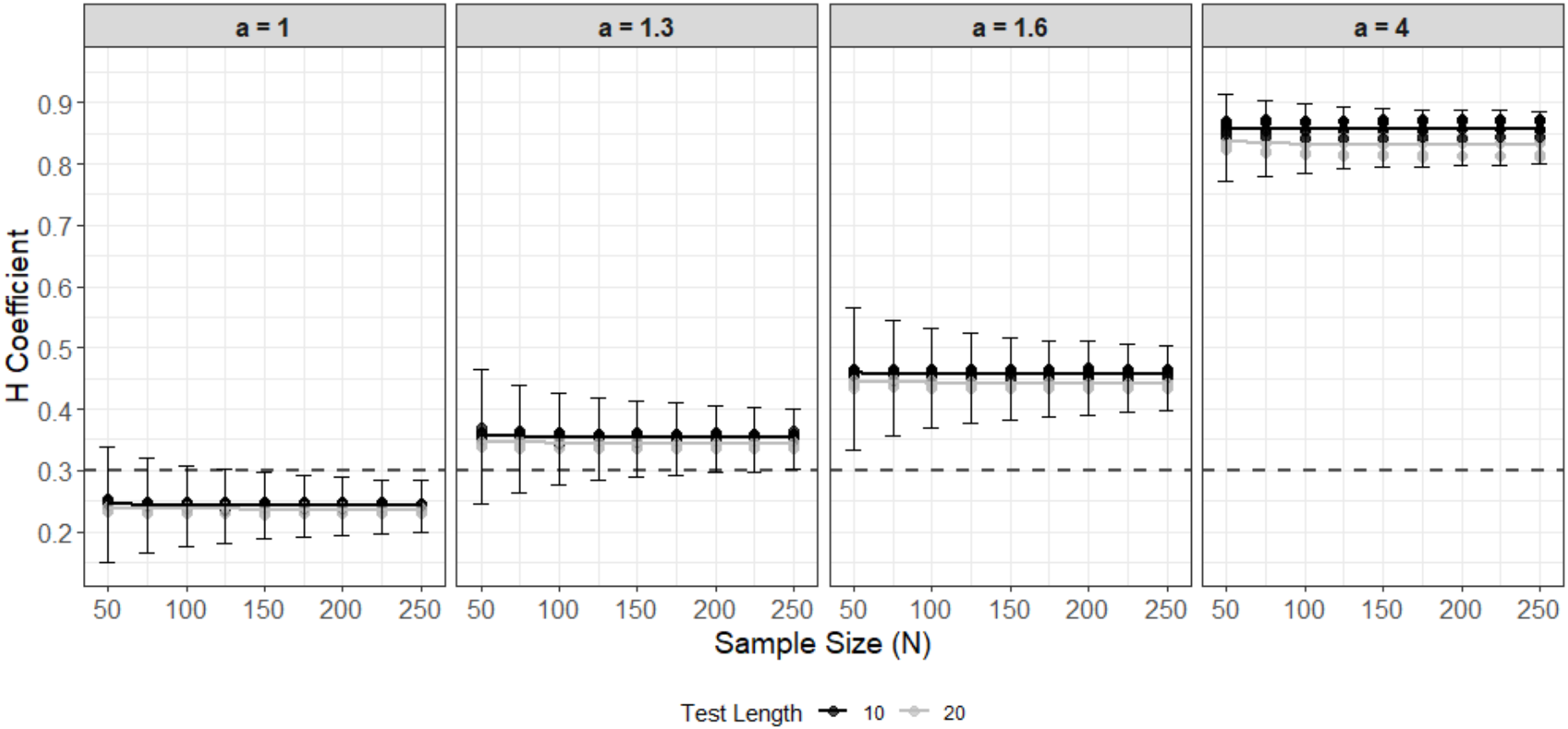
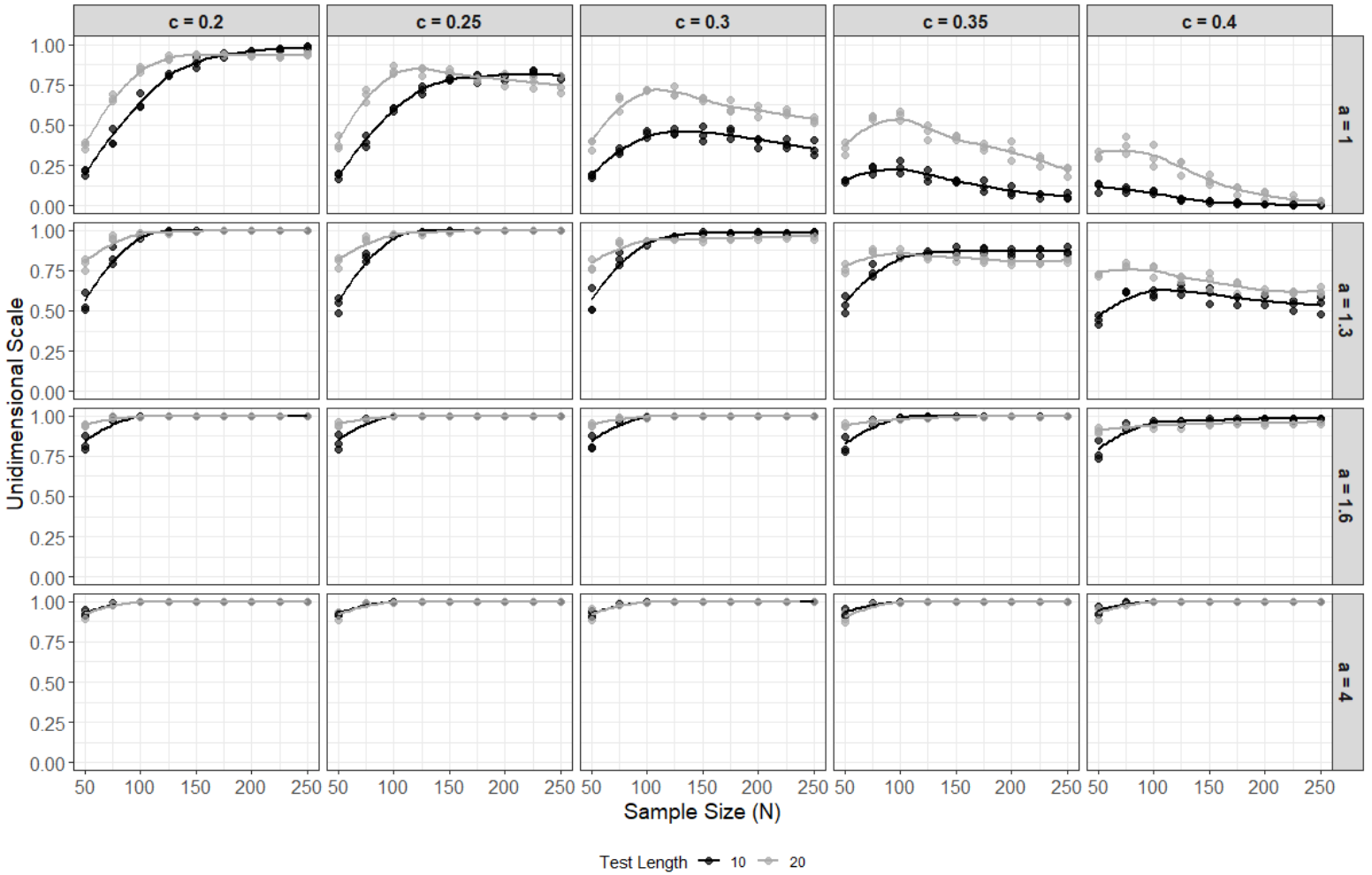


Figure 2. Frequency of *Unidimensional Scale Retrieval versus Sample Size for Different Levels of Item Discrimination and Lower Bound*

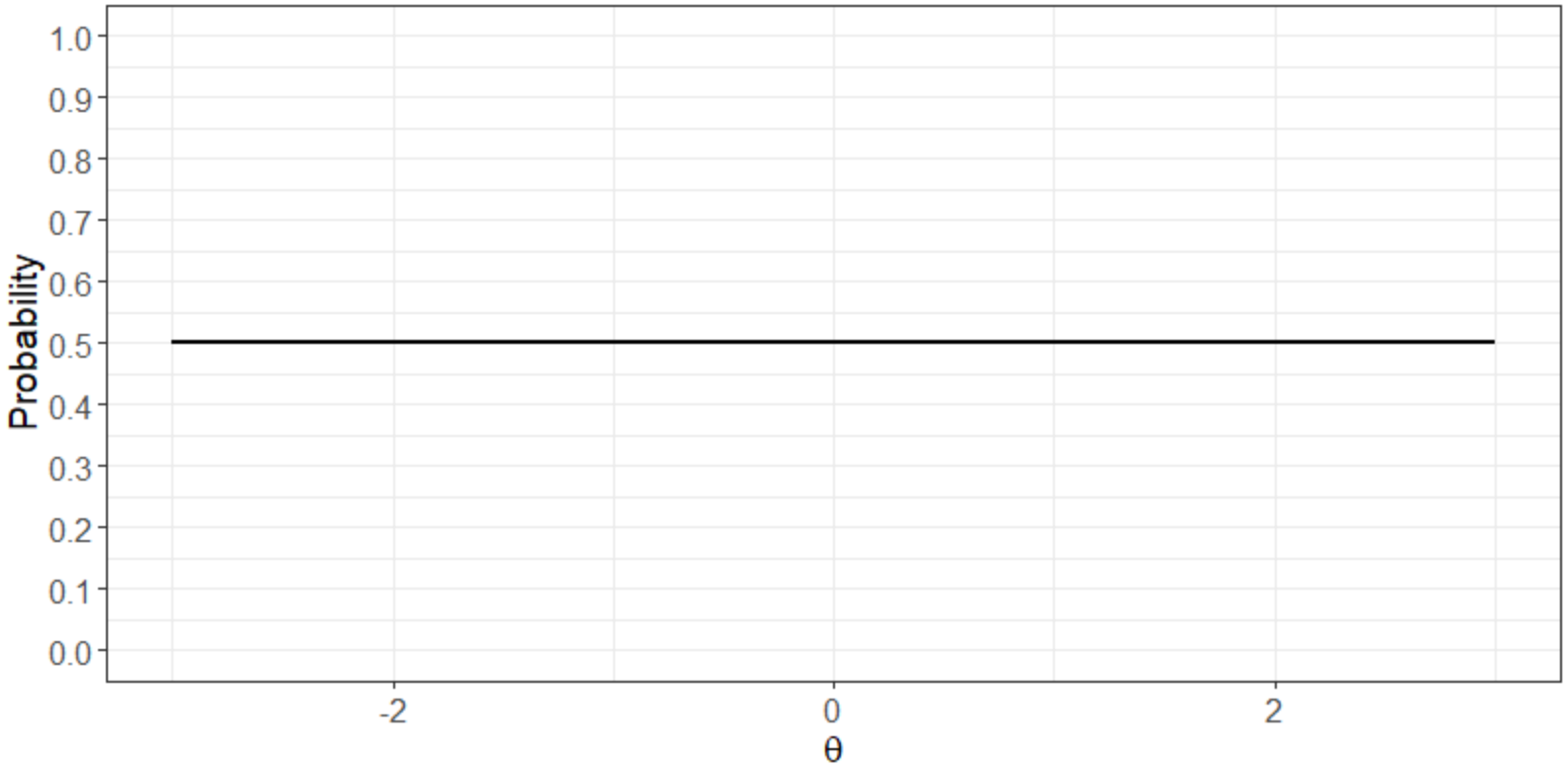


Deviant Items

› Replacement of items before applying MSA

Type	Number	Location	Test Length
binomial	*ndev* = 1	b_5	$J = 10$
binomial	*ndev* = 3	b_3, b_5, b_7	$J = 10$
PARELLA (Hoijtink, 1991)	*ndev* = 1	b_5	$J = 10$
PARELLA (Hoijtink, 1991)	*ndev* = 3	b_3, b_5, b_7	$J = 10$

Deviant Items: *Binomial*



Deviant Items: *PARELLA*

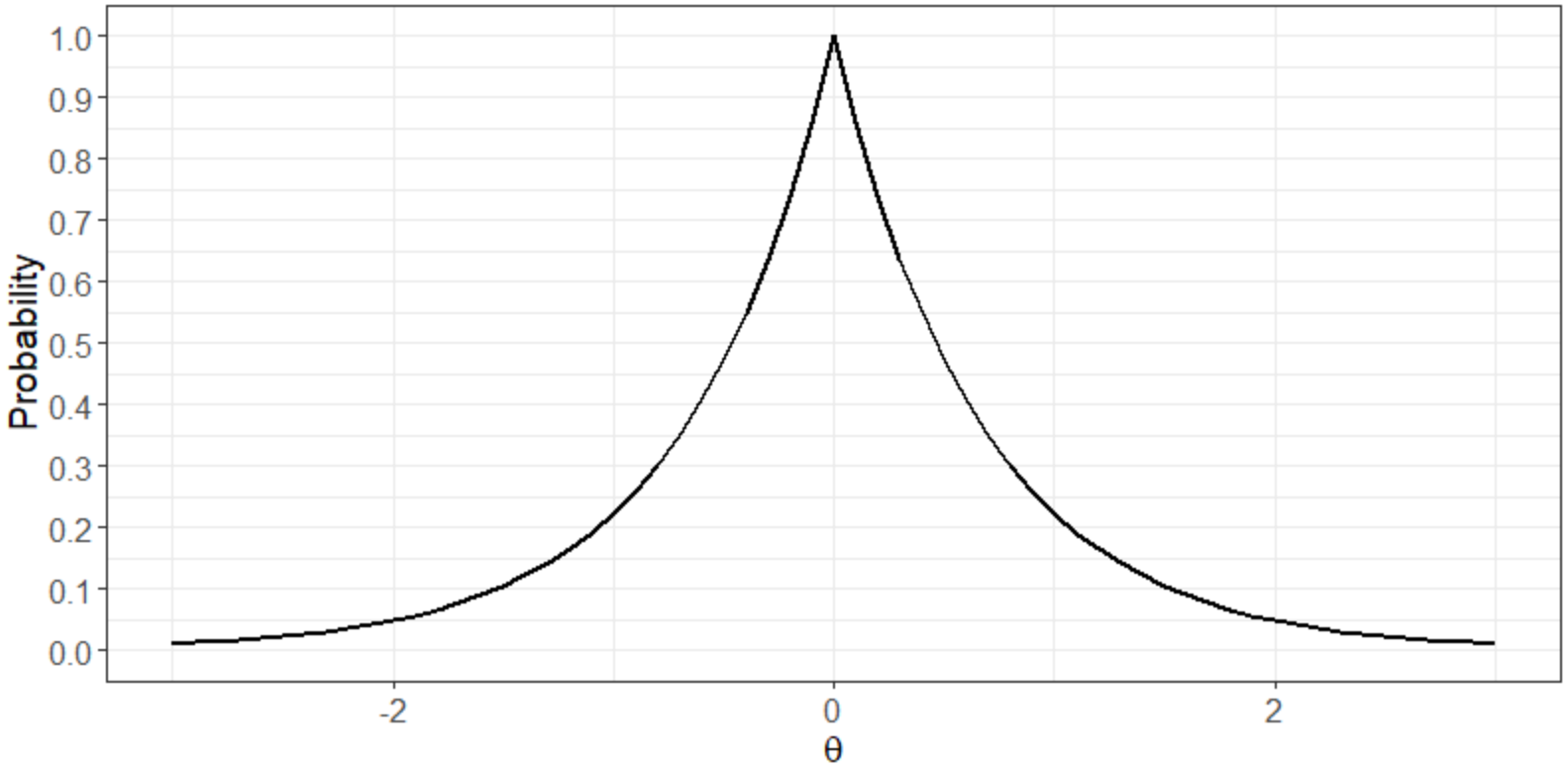


Figure 3. *H Coefficient versus Sample Size if Deviant Items are Present for Test Length J = 10*

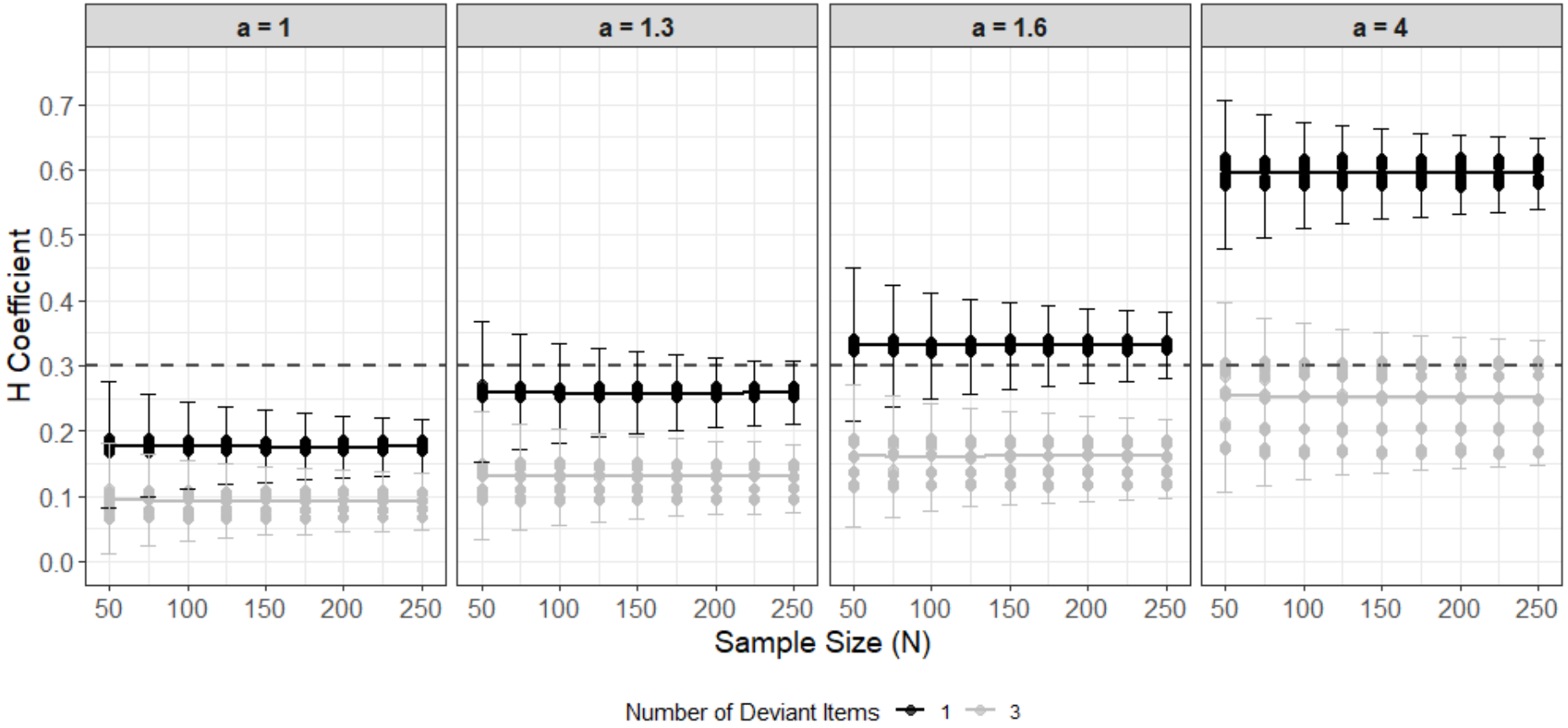
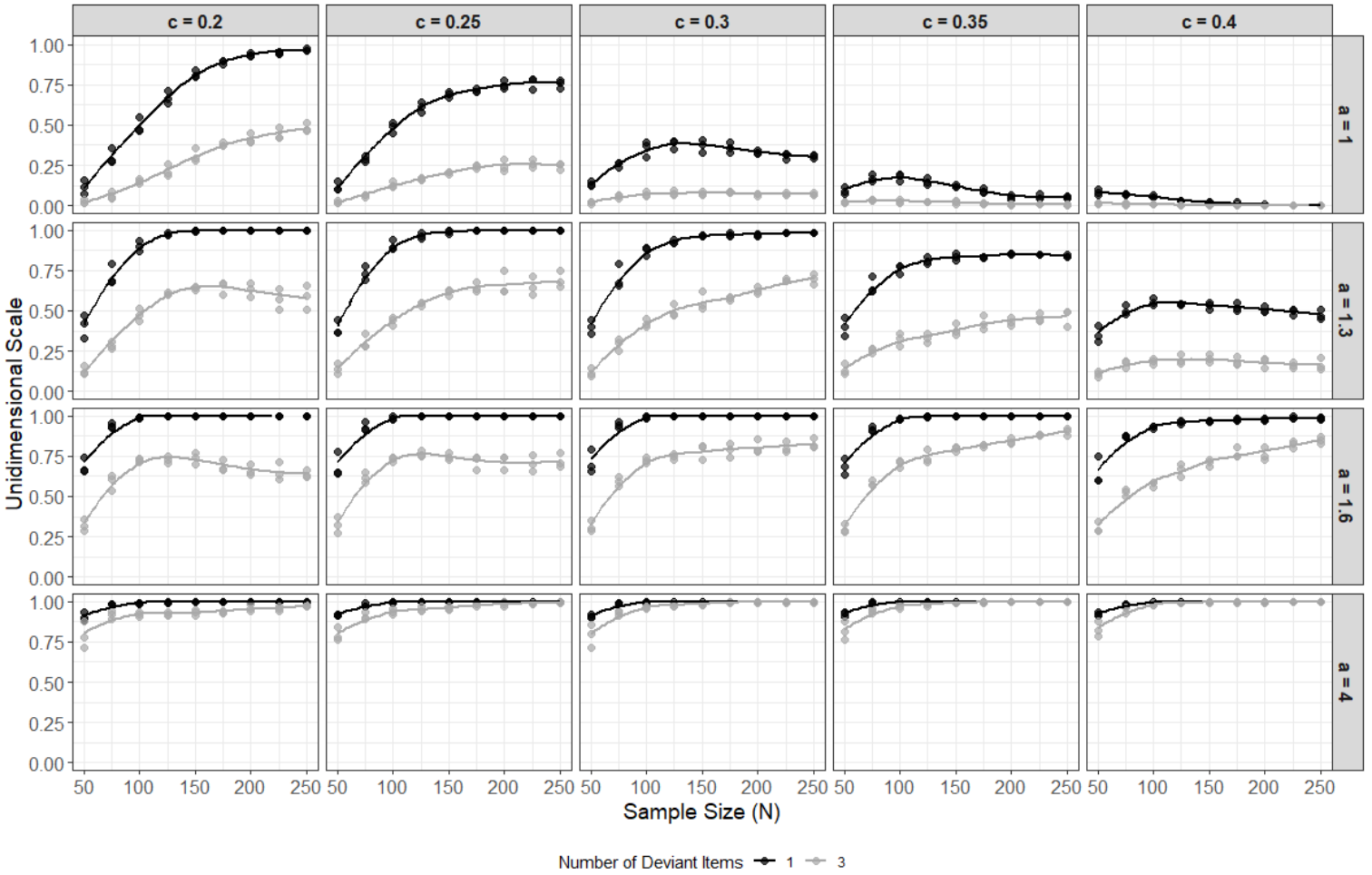


Figure 4. Frequency of *Unidimensional Scale Retrieval versus Sample Size for Different Levels of Item Discrimination and Lower Bound if Binomial Items are Present. Test length J = 10.*



Discussion & Conclusion

- **Sample Size & Exploratory MSA Performance**:

- Increasing sample size substantially enhances the likelihood of scale retrieval if item discrimination is *sufficiently* large

Discussion & Conclusion

- **Item Discrimination & Lower Bounds**:

- Adequate item discrimination a ≥ 1.3 (correspond to *H* > .3) is essential for robust unidimensional scale solutions, especially with small samples

- Flexible lower bounds could improve exploratory MSA in low-discrimination and small sample contexts

Discussion & Conclusion

- **Impact of Deviant Items**:

- The presence of multiple deviant items substantially decreases *H* coefficients and scale retrieval rates

Thank you for your attention!

Questions and Comments

?

Literature

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Additional Slides

Methods: AISP

- *Mokken* in Rstudio (van der Ark, 2012)
- Iterative approach to construct Mokken scales (van der Ark, 2007)

- If ($H_{jk} > 0_{(sig.)}$ & $H_{jk} \geq c$) → candidate!
  - Which candidates are max(H_{jk}) → scaled!
- 3rd items: if($H_j > 0_{(sig.)}$ & $H_j \geq c$) → candidate!
  - Which candidates is max(H) → scaled!
- … if no more candidates
- Rest items: if ($H_{jk} > 0_{(sig.)}$ & $H_{jk} \geq c$) → 2nd scale,…
  - else → items unscalable!

Methods: Item Difficulty Pattern

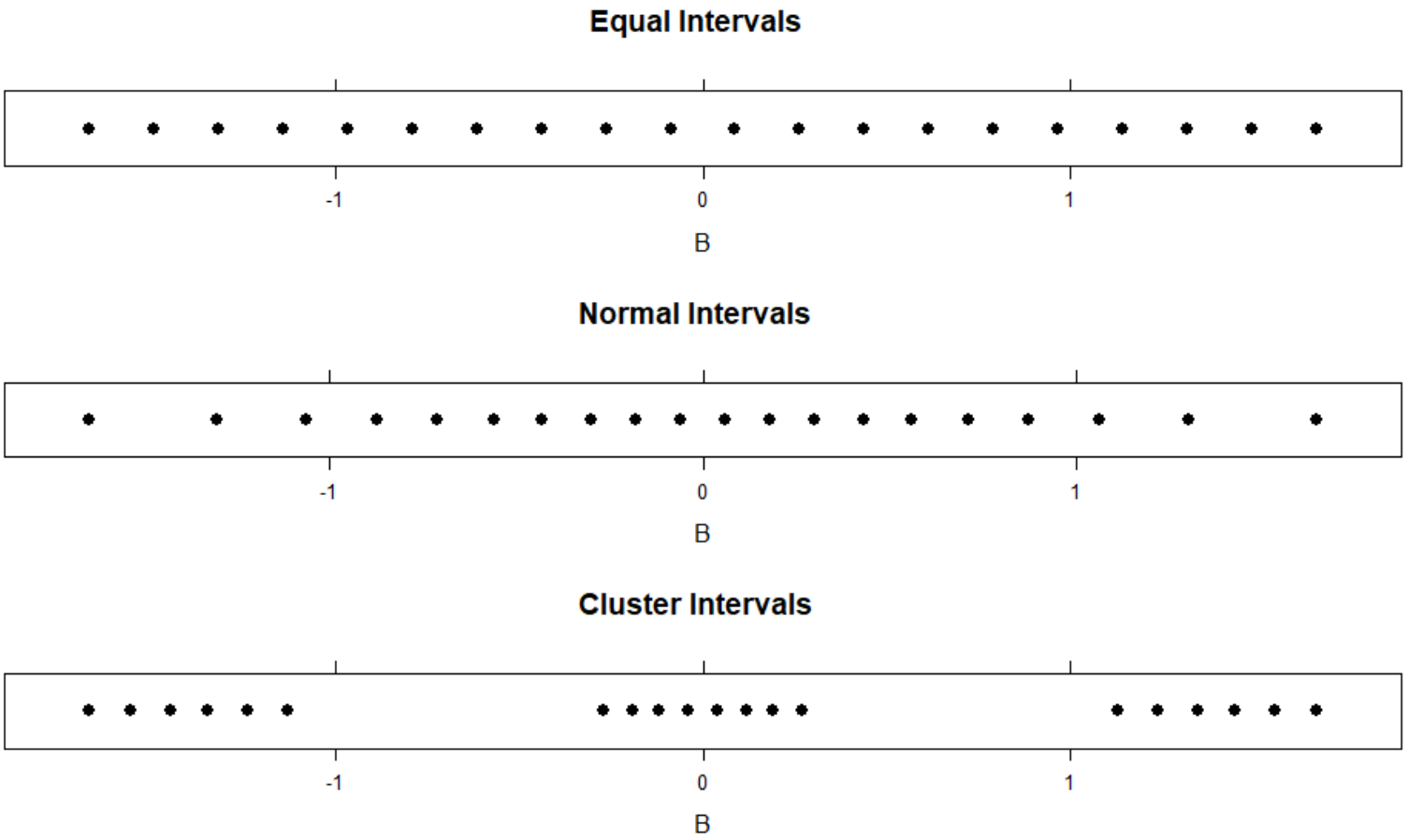
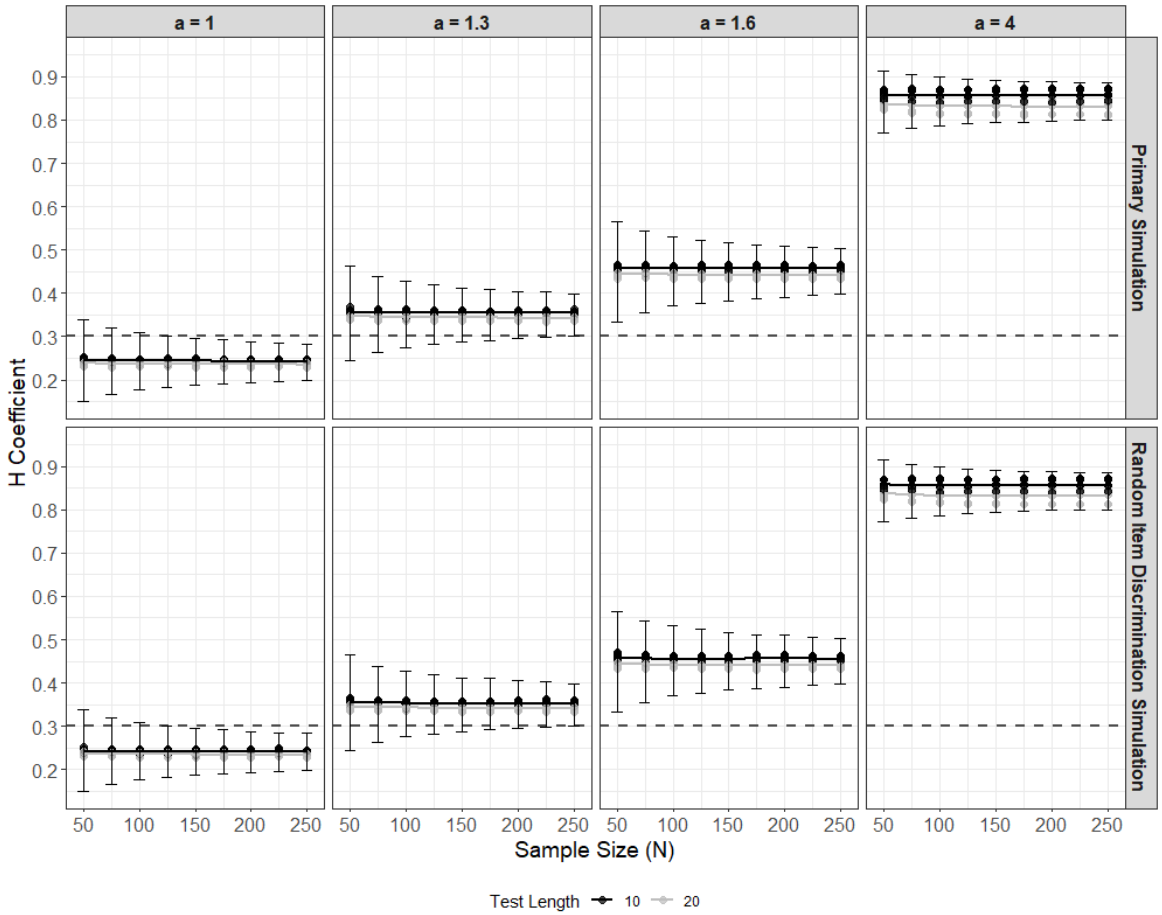
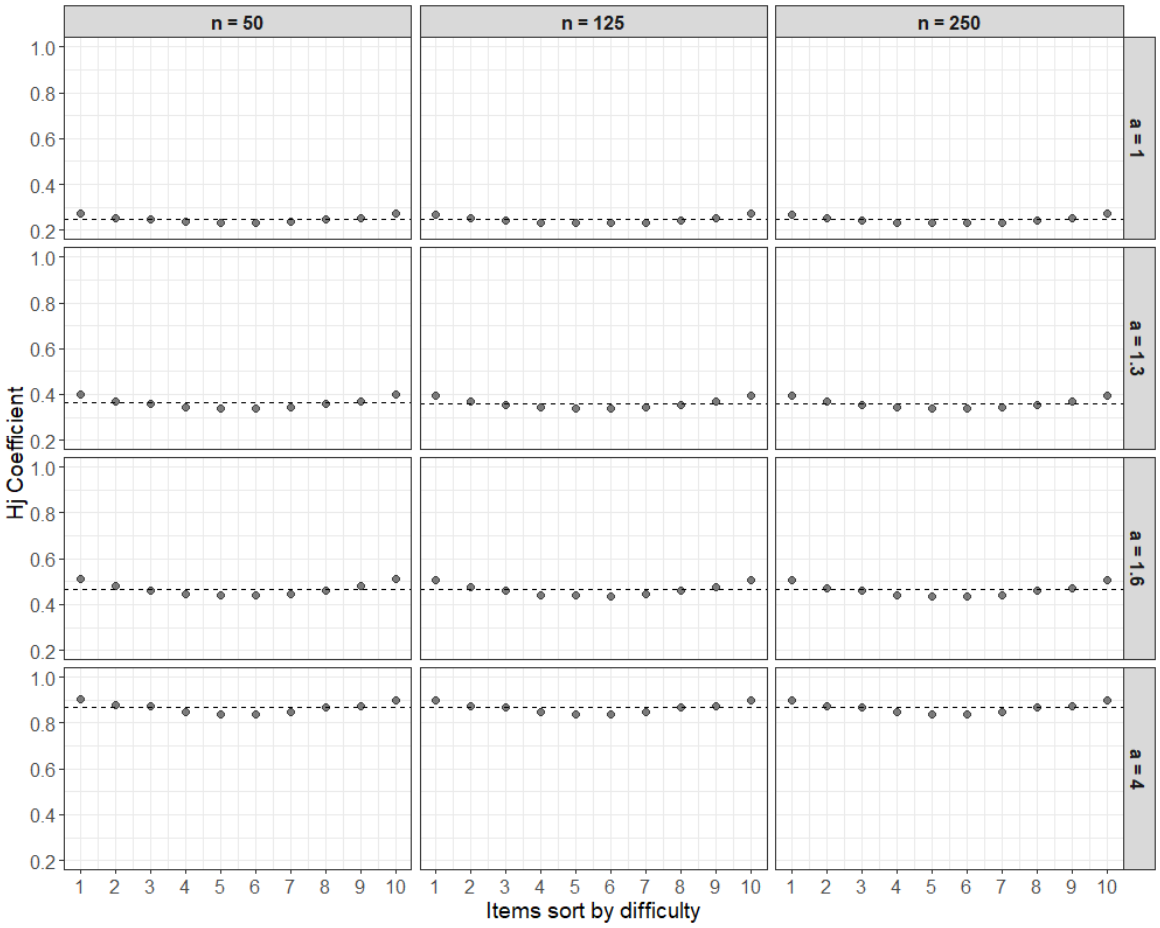


Figure. *H Coefficient versus Sample Size for Different Levels of Item Discrimination split by Simulation Trials*



Figure

Hj Coefficient versus Items sort by Difficulty and split by Sample Size and Item Discrimination



Figure

Standard Deviation (Hj) versus Items sort by Difficulty and split by Sample Size and Item Discrimination

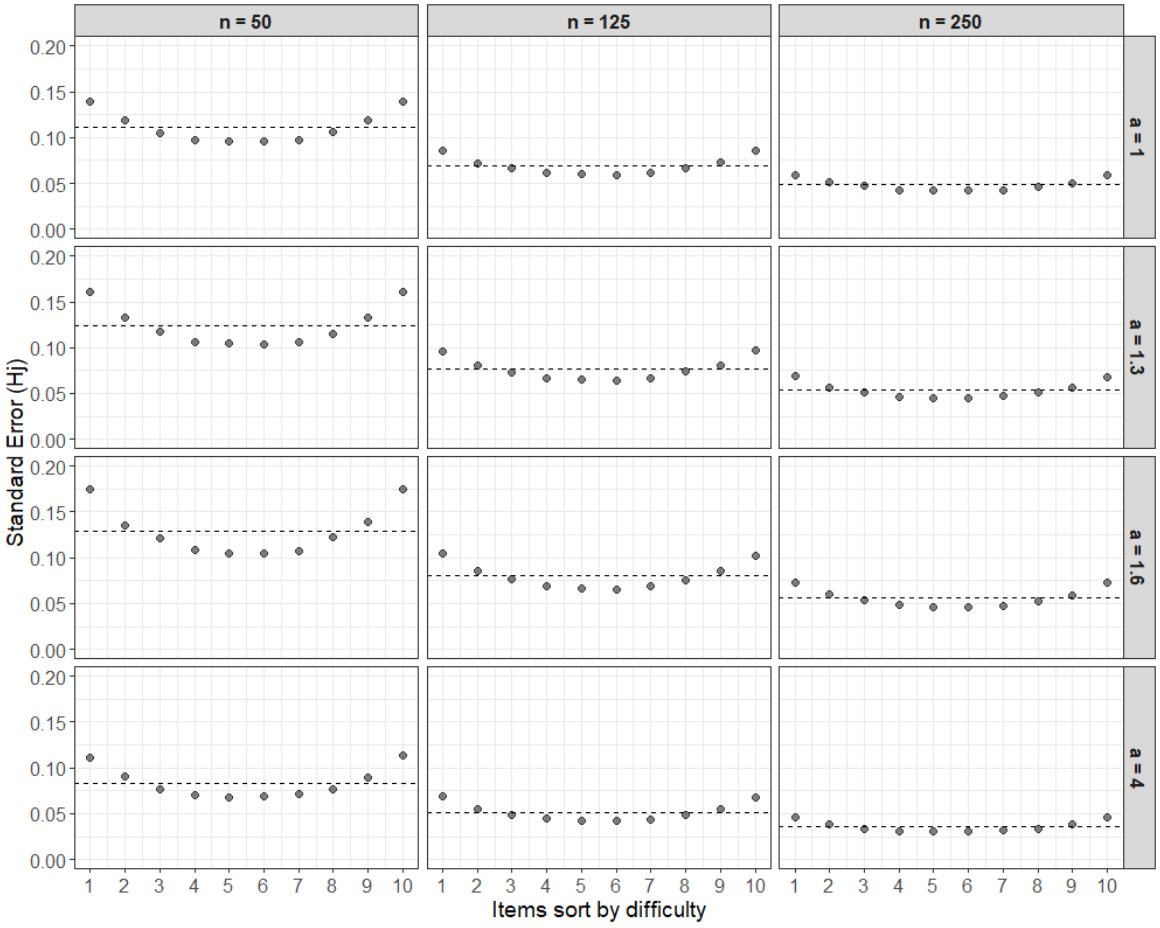
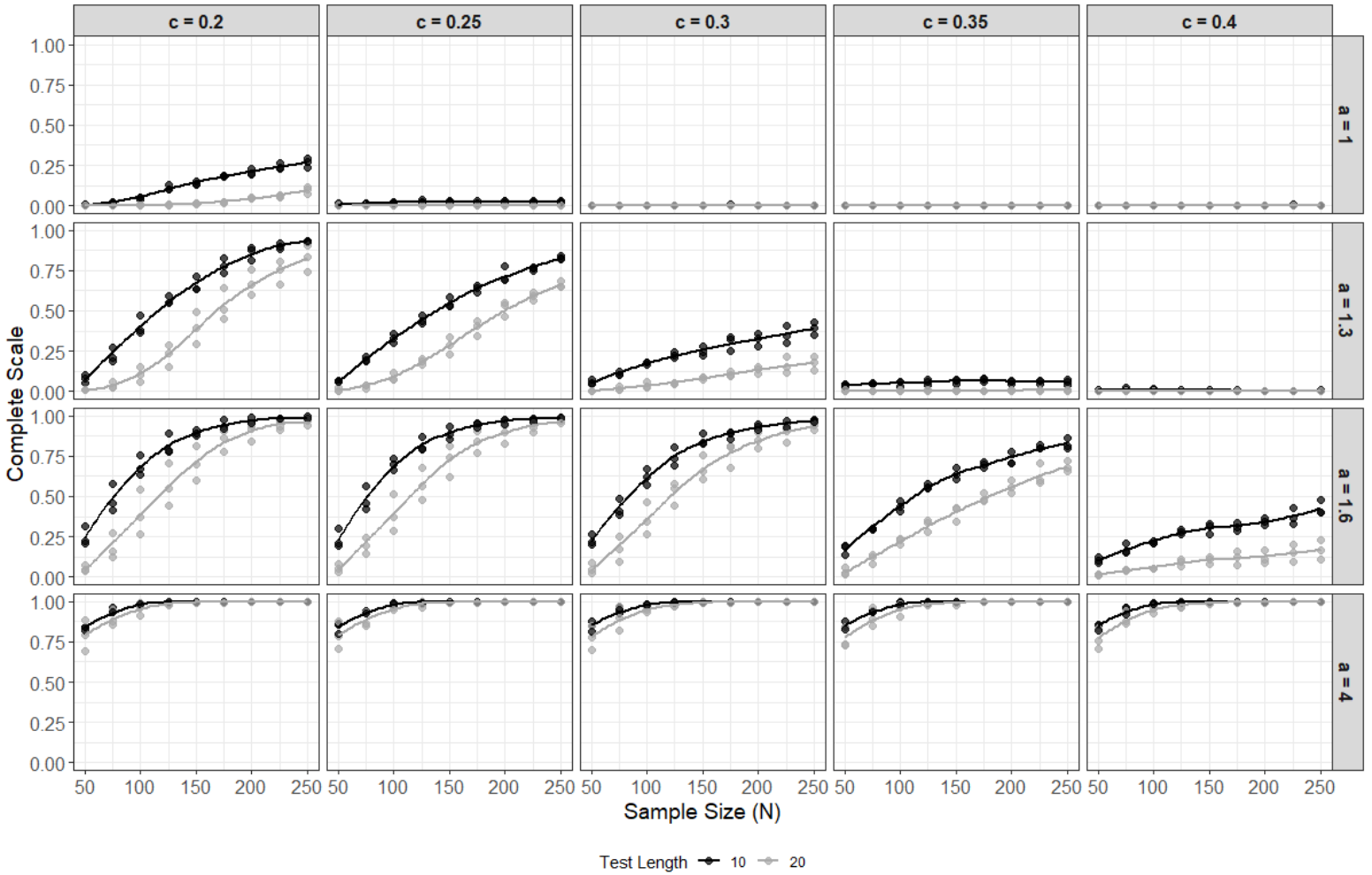
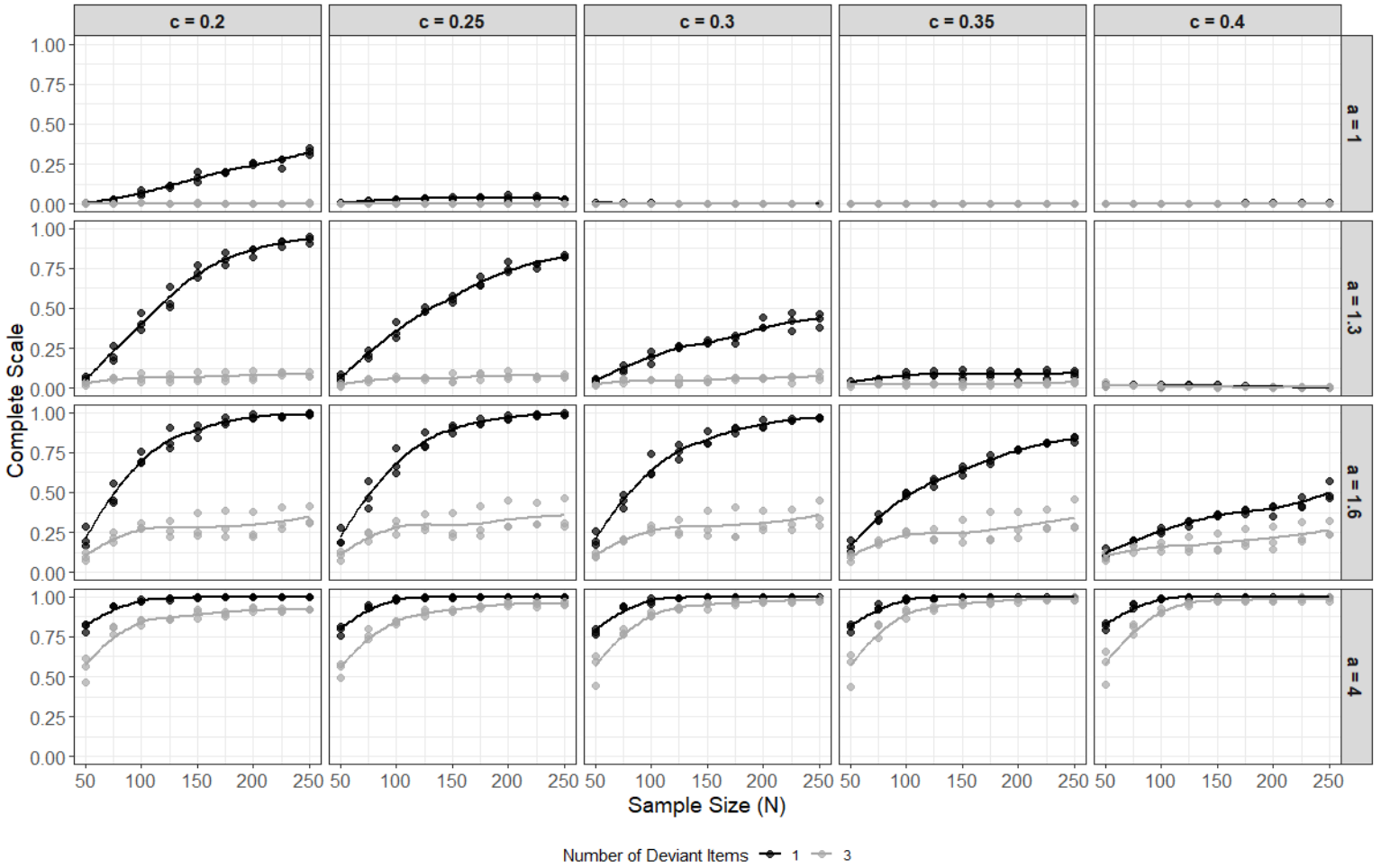


Figure. Frequency of *Complete Scale Retrieval versus Sample Size for Different Levels of Item Discrimination and Lower Bound*



Source: Main Simulation Trial

Figure. Frequency of *Complete Scale Retrieval versus Sample Size for Different Levels of Item Discrimination and Lower Bound if Binomial Items are Present. Test length J = 10.*



Source: Deviant Item Simulation Trial

Figure. Frequency of *Complete Scale Retrieval versus Sample Size for Different Levels of Item Discrimination and Lower Bound if PARELLA Items are Present. Test length J = 10.*

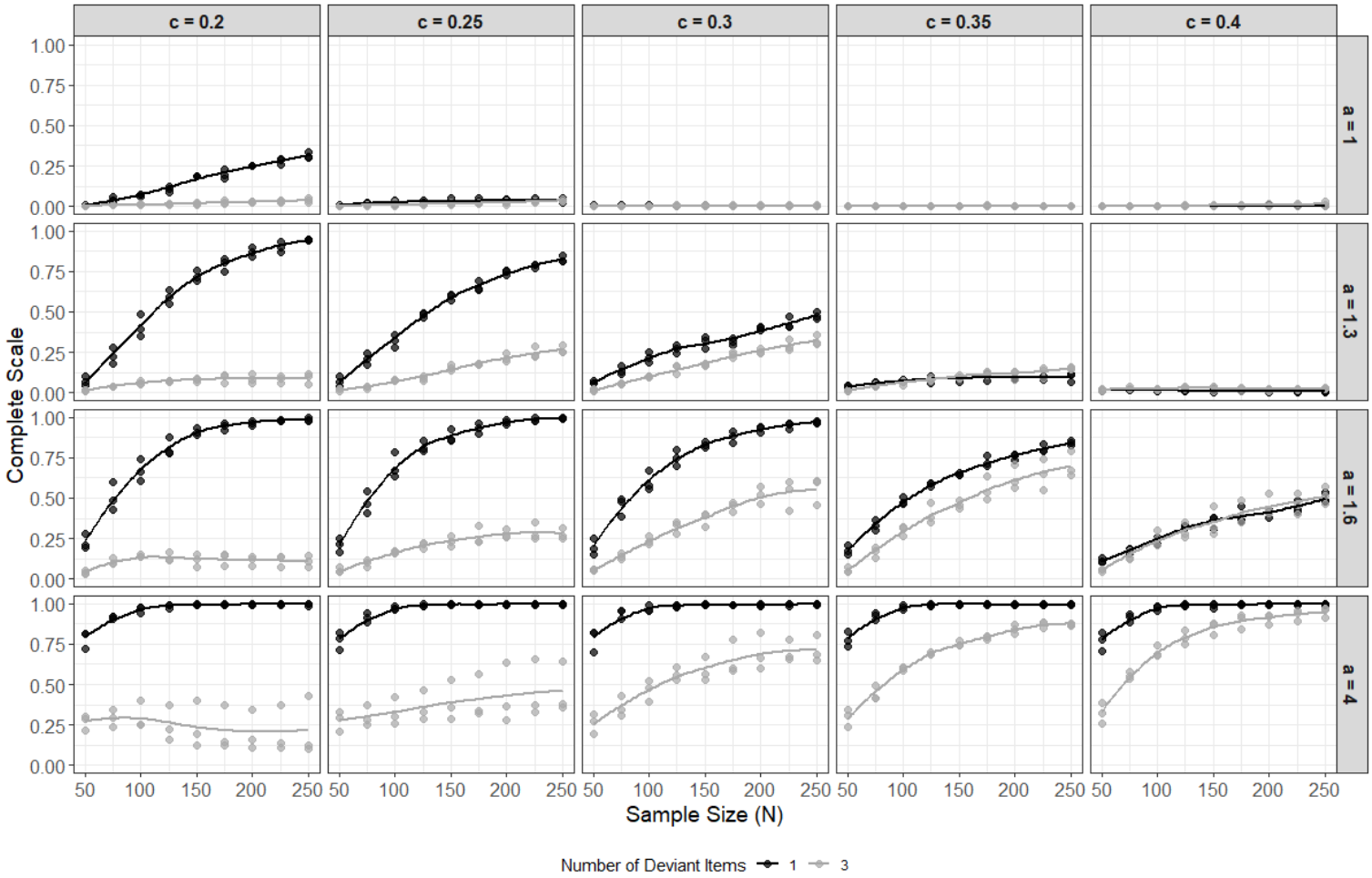
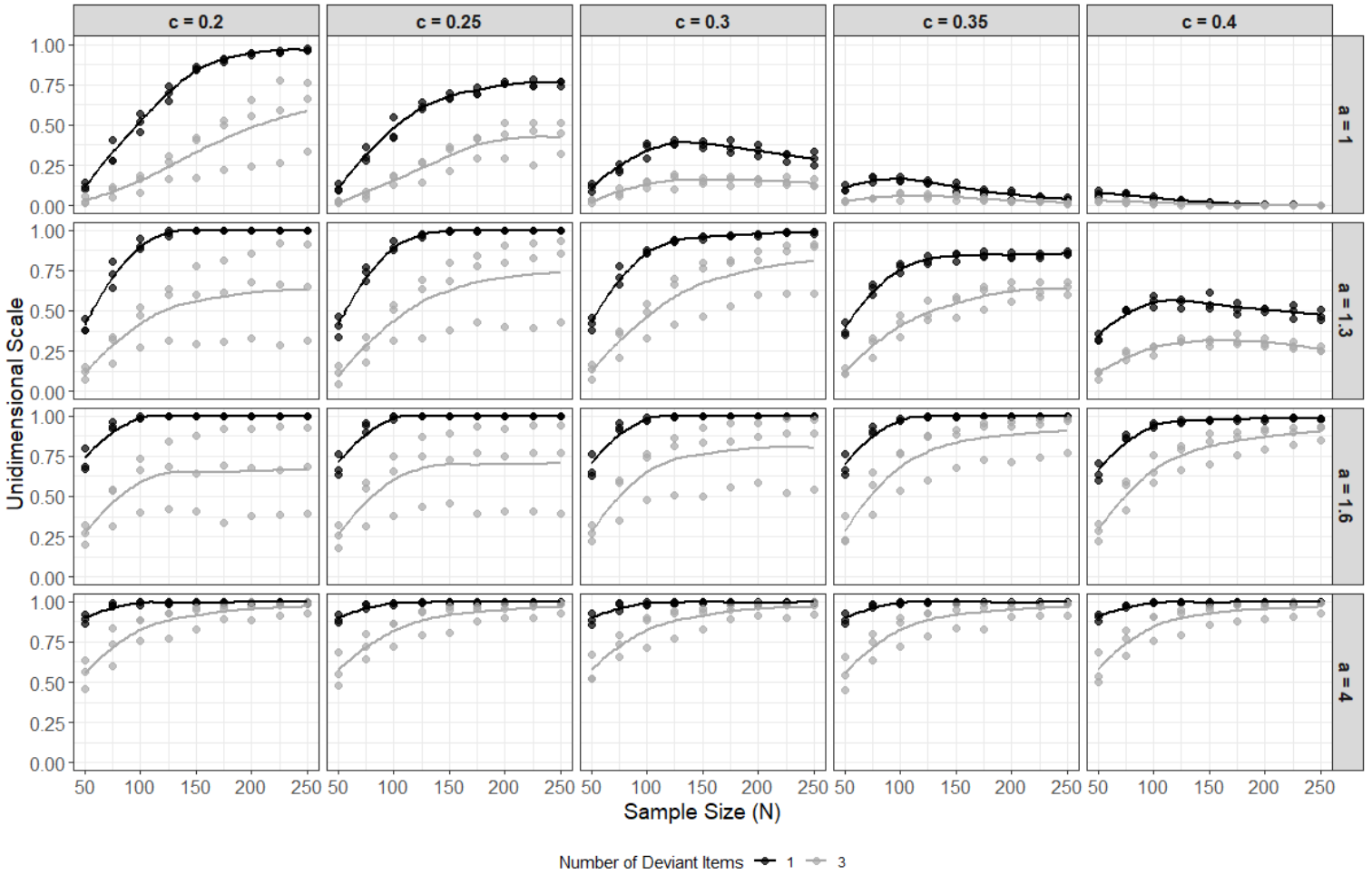


Figure. Frequency of *Unidimensional Scale Retrieval versus Sample Size for Different Levels of Item Discrimination and Lower Bound if PARELLA Items are Present. Test length J = 10.*



Discussion & Conclusion

- **Confirmatory MSA Findings**:
- Item discrimination is key
- Extreme item difficulty relates to larger H_j-coefficients (Kuijpers et al., 2016)